

A SIMPLE QUANTITATIVE TRADING EXAMPLE: SPREADS

1. WORKING OUT A SPREAD TRADE

Let's work out what the bones of a quantitative trading strategy might entail by examining one of the simplest examples: the spread trade.

Spread trading strategies have a long history and many current forms. Because they involve trading more than one asset, they are just about the simplest example of a case where we can stochastically control risk versus reward.

1.1. What Is A Spread? Let's say we monitor prices or returns on a pair of securities. For sake of argument, we will say we are monitoring the prices $f_t^{(2)}, f_t^{(5)}$ of 2 and 5 year CME treasury note futures ¹. We can construct a new variable consisting of the difference between them

$$s_t := f_t^{(5)} - f_t^{(2)}$$

This value s_t is called the *spread* or *difference spread* between the 2 and 5 year note futures, and we often think in terms of trading spreads, even when in practice the trades are expressed in underlying securities such as futures contracts.

1.1.1. Return Spreads. For assets with clear fundamental relations, we tend to use difference spreads as above. Frequently, we prefer to assume that old differences are "priced in" and therefore unlikely to change, in which case we may prefer *return spreads* in which we consider how much returns for the two assets have differed in recent history.

1.2. Reversion To The Mean. We might find that, during the 20 years from 1995 to 2015, the average of s_t was $\bar{s} = 8.5$. A strategy would plausibly make money by always making bets that, whatever the value of s_t , it is expected to revert back to around 8.5. How would such a spread trading strategy look?

It probably does not make sense to make bets when the current s_t is very close to 8.5. So, for example, if $s_t = 8.499$ or $s_t = 8.501$ we should not hold a position. On the other hand, if $s_t = 4.0$ this looks like a good opportunity for us to bet that s will soon rise. We do not necessarily know if $f_t^{(2)}$ will fall or $f_t^{(5)}$ will rise, but we think some combination of those things will happen. So it makes sense to both short some 2 year note futures and buy some 5 year note futures. We call this "buying" the spread because we are trading securities in such a way that we think the spread will rise.

For similar reasons, if we see $s_t = 12.0$ then we ought to buy some 2 year note futures and short some 5 year note futures. We call this "shorting" the spread.

Now, if $s_t = 14.0$, it is easy to argue that the opportunity is greater and we should be shorting even more of the spread. But remember, it probably only got to 14.0 by going through 12.0 at some point. We are likely *already* short the spread and, since it has risen further, have lost money on the position. Are we prepared to risk losing more?

¹We will ignore details of futures and bond fractional pricing in this discussion.

In spread trading, it is common to have some point at which you admit that your hypothesis (of spread reverting to its mean) has been so contradicted by market data that you are no longer willing to believe it, at least not with your dollars. Such a point is often defined in terms of PL – dollars lost to the position, and is called a *stop loss level*. Once you have reached it, you resolve to close out all positions, and perhaps wait a while before attempting a similar spread trade again.

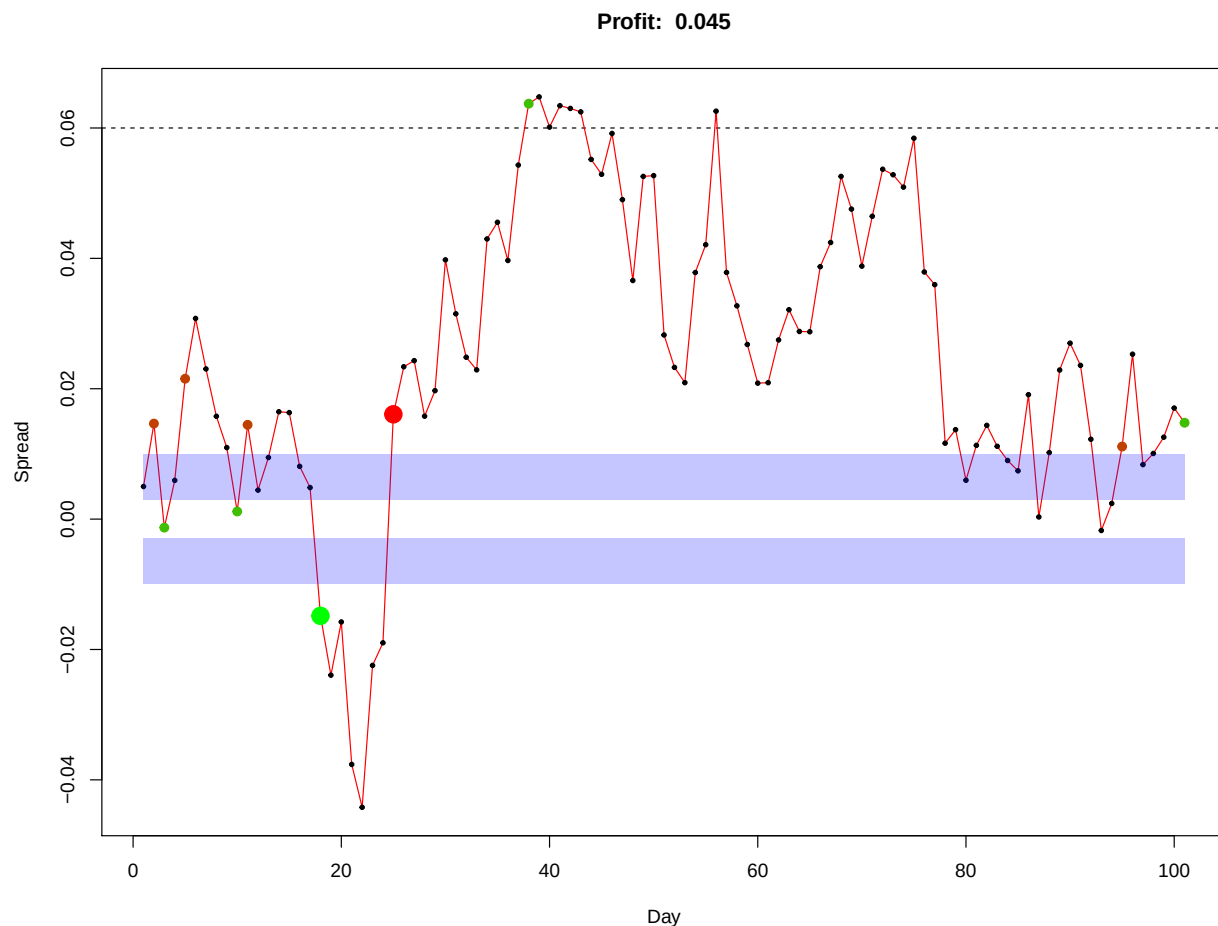


FIGURE 1. Trading A Hypothetical Spread

The *human* complexity of writing code for spread trading is very low. It can be done in 50-150 lines of R or Python, and is feasible even in Excel. However, when we consider how we might approach spread trading in *practice*, we find there are many parameters to determine and they are likely to make us run simulations over and over. These parameters include

- Which security pairs to run on
- Entry and exit boundaries
- Hedge proportions, if not 1:1
- Hedge determination algorithms
- Lookback periods for hedge (and possibly returns) computations
- Stop loss/ position sizing boundaries

Testing 16 possibilities for each of the above would introduce high computational complexity in the research or optimization phase. An exhaustive grid search would lead to running over 15 million backtests. If each one takes 5 seconds, we will need about 3 years to run them all. We can consider this research or optimization complexity.

1.3. Generalizations To More Stochastic Control. Though the (nominally) zero-centered series we have based our scheme on was a direct spread, we can easily imagine first running a regression of future returns against current observed spread. This crude estimate of *alpha* can then serve as the basis on which we choose to send orders to the market in a very simple trading scheme.

At this basic level, trading on alpha looks very much like spread trading. We have entry and exit criteria, position tracking and stop losses and so on. One difference is that, for most models, alpha takes persistence into account. So alpha tends to change over time even if little or nothing changes in the inputs.

1.4. Other Major Considerations. Our little vignette has left some major considerations out of the picture. Most importantly, in the real world there is never "a price" for $f_t^{(2,5)}$. There are bids and offers, and anyone wanting to effect a trade immediately must either pay offers or hit bids, in which case we say we are sending aggressive *market orders* or more likely *limit orders* with a *limit price* "through the book". Alternatively, we can enter our own bids and offers into the *order book* in the hopes that someone else will send an aggressive order with us at some unknown future time.

If our scheme is controlling market orders, we may well assume that (given a cost model) we are able to take on a small position with relative certainty. If instead the scheme is controlling *resting orders* (i.e. *liquidity-adding* or *passive* orders) then we must further account for tracking outstanding orders and when to cancel them. This does not apply only to electronic markets. Over-the-counter markets have human equivalents (often called *standing* orders) to the passive electronic orders.